

Francisco Rau

Football Data Analyst - Recruitment & Player Valuation

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Summary

Data analyst with an MSc in Data Science & Society and a BSc in Economics (Tilburg University), building recruitment analytics as my own research programme. transfer-alpha is a player-valuation and recruitment platform I built end to end - 163 competitions, 44,000+ player-seasons, a fair-value model and calibrated 12-month probabilities that a player's market value moves. Professionally I spent a year turning operational data into production decision tooling. I care most about work that survives its own audit: the numbers below include the ones that got worse when I checked them properly.

Selected Football Work

transfer-alpha - Player Valuation & Recruitment Platform 2026 - Python, scikit-learn, statistical validation

- Built the full pipeline solo: harvesting per-match ratings and market-value histories, entity resolution across sources, feature construction, modelling, calibration and a browser front end. Coverage: 163 competitions, 44,403 player-season rows.
- Fair-value model estimates what a player's market value should be from performance, age, minutes, league quality and club context (R^2 0.59; 65% directional accuracy against a 57% base rate). The gap between fair value and market price is the scouting signal.
- Calibrated 12-month probabilities per player - $P(\text{value rises})$, $P(\text{falls})$, $P(\text{rises } 50\%+)$ - from a gradient-boosting classifier with isotonic calibration. Rolling out-of-sample design (train to y-2, calibrate on y-1, test on y): AUC 0.87-0.90, Brier skill score +0.33 to +0.48, clean reliability curves.
- Every probability ships alongside its twin computed WITHOUT the valuation gap, so a card can state honestly when a call is driven by age and price rather than by the model's signal.
- Leak-audited backtest of transfer outcomes: shortlisted players were sold at 49.7% vs 30.2% for the comparable pool (1.65x), while sale multiples were 1.11x vs 1.06x - i.e. the edge is in identifying who moves, NOT in selling higher. An earlier version reported 3x better results; auditing it found look-ahead bias in how prices and season vintages were dated, and I rebuilt both.
- Tested an additional data provider and dropped it: forward testing showed no predictive gain (IC 0.359 without vs 0.352 with) across five variants, so it stayed out of the model.

Player Market-Value Modelling

2024 - Python, statistical modelling

- Predicted footballers' market values across a multi-year, 100+ competition dataset I compiled myself. Best model: R^2 0.921, RMSE \$2.23M.

Experience

Data & Process Analyst - Netvice

May 2025 - Jul 2026 - Tilburg, Netherlands

- Established operational BI as a new, self-owned workstream - a first in the company's ten-year history - owning it from roadmap to daily use on the floor.
- Forecasting model turning planned volume into the exact headcount needed on any future day, 500+ days ahead, with capacity-choke detection and a batch-splitting optimiser.
- Percentile-based pacing metric making individuals comparable across campaigns with very different workloads, combined with outcomes into one performance index - the same normalise-then-rank problem as comparing players across leagues.
- Fully self-directed: each system began as my own proposal - need identified, pitched, built and shipped.

Student Researcher - Philips

Sep 2022 - Dec 2022 - Eindhoven, Netherlands

- Built linear models estimating adoption of a subscription service across student segments.

Skills

Modelling: gradient boosting, regression, probability calibration (isotonic), rolling out-of-sample validation,

Brier/AUC/reliability, information coefficients, look-ahead auditing
Engineering: Python (pandas, scikit-learn), data harvesting & entity resolution across sources, PostgreSQL,
TypeScript/React front ends, Git, AI/agent dev tooling
Domain: player valuation, transfer-market dynamics, squad and contract context, league-strength adjustment

Education

MSc Data Science & Society (Business Track) - Tilburg University	2023 - 2024
BSc Economics - Tilburg University	2020 - 2023

Languages

English (native), Spanish (native), Dutch (A2, actively learning)